



## Link List

1. What is linked list.
2. Implementation Of linklist.
3. Node class
4. Linklist class .
5. Creation of linked list

A linked list is a sequence of data structures, which are connected together via links. Linked List is a sequence of links which contains items. Each link contains a connection to another link. Linked list is the second most-used data structure after array.

Arrays demand a contiguous memory location. Lengthening an array is not possible. We would have to copy the whole array to some bigger memory location to lengthen its size. Similarly inserting or deleting an element causes the elements to shift right and left, respectively.

But linked lists are stored in a non-contiguous memory location. To add a new element, we just have to create a node somewhere in the memory and get it pointed by the previous element. And deleting an element is just as easy as that. We just have to skip pointing to that particular node. Lengthening a linked list is not a big deal.

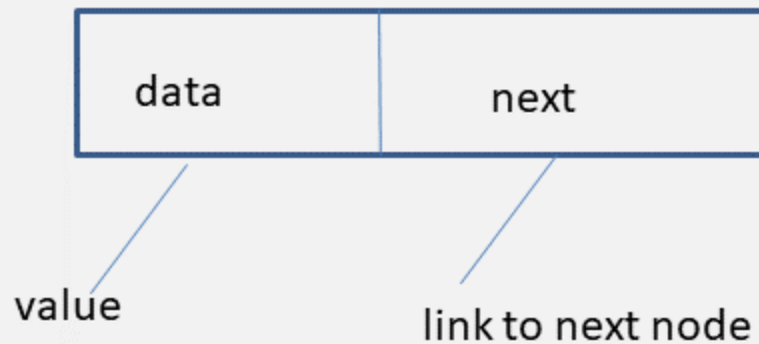
## 1. Disadvantages of Arrays

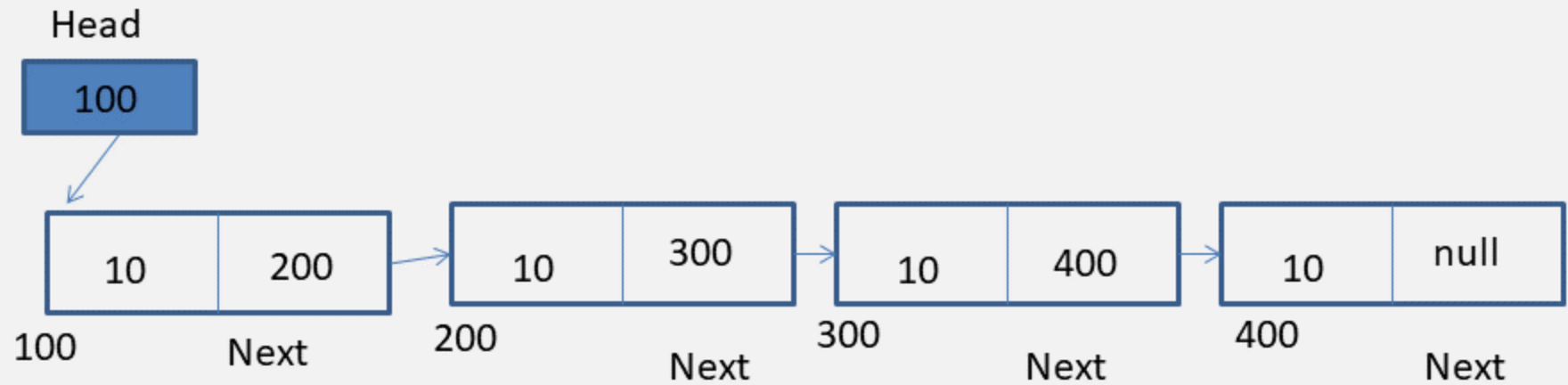
- slow Searching in unorders array.
- Insertion and deletions are slow.Because, we have to shift subsequent elements.
- Fixed size.
- Wastage of memory.

## 2 Linked list

- linked list is able to grow in size as needed
- does not required the shifting of items during insertion or deletion

1. Linked list is linear datstructure that support the dunamic memory allocation. (The amount of memory could be varied during it's use).
2. Linked list is used to hold sequence of data values.
3. Data values need not be stored in adjacent memory cells.
4. Each data values has link which indicate where its next. data value in computer memory.
5. An element in a linked list is known as a node. node contains data part and link part.





```
class Node
{
    private int data;
    private Node next;
    public Node()
    {
        data=0;
        next=null;
    }
    public Node(int data)
    {
        this.data=data;
        this.next =null;
    }
    ..... Set and get methods..
}
```

```
class linklist
{
    private Node head;
    public head()
    {
        head=null;
    }
    public void createLinklist(int numberOfTerms)
    {
        .....
    }
    public void display()
    {
        .....
    }
}
```